

ColorClock

Tell time through color

Introduction

What is ColorClock?

Introduction

Color Foundations

System Overview

Time Controller

Light Display

User Interface

Circuit Board Evolution

Deployment

Final Thoughts

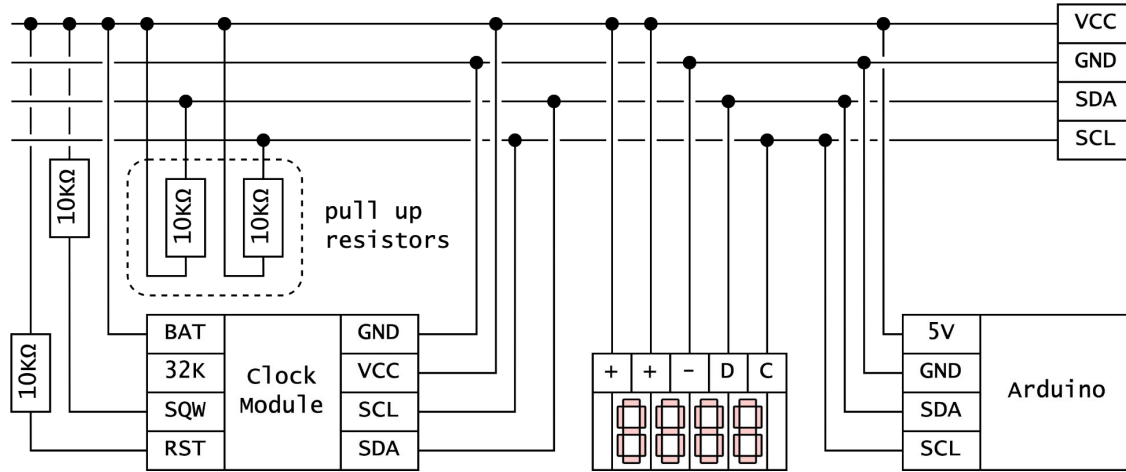


Software Prototype 2020

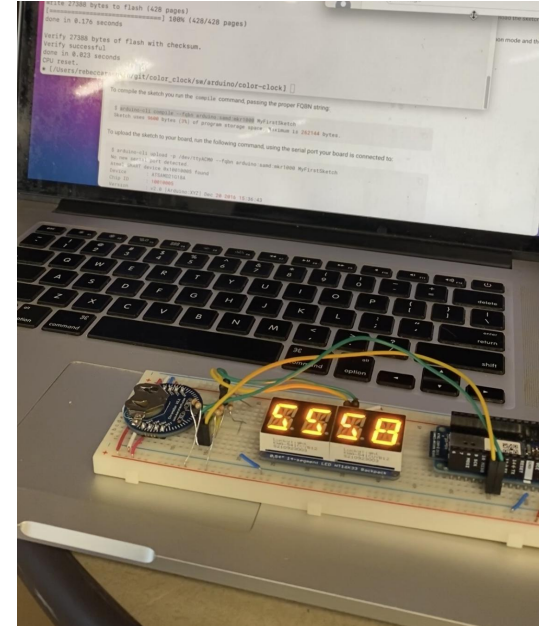
- Built with Processing:
Java-based language
for digital art
- Exercises color-time
mapping algorithm
- Cycle time: 6 seconds



Module Prototyping 2022



Circuit Diagram
May 2022



Color Foundations

Mixing color with pigment
and light

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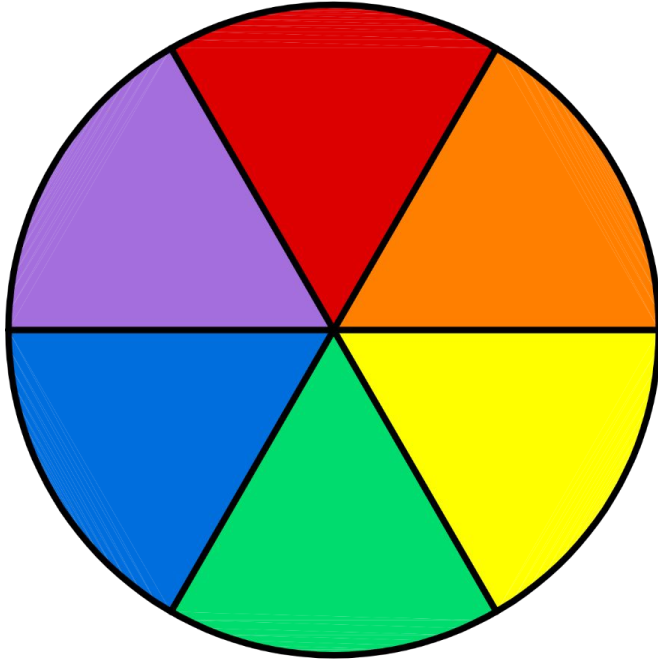
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Pigment Primary Colors: Red, Yellow, Blue



Red

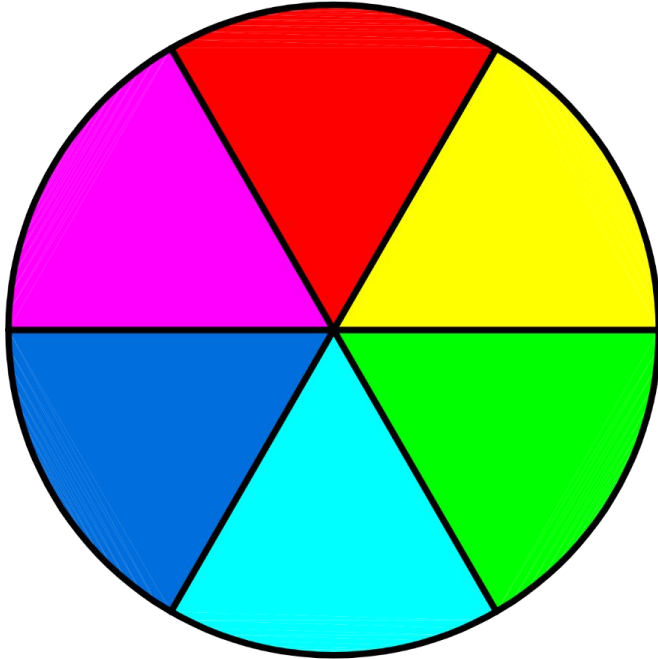
Yellow

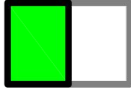
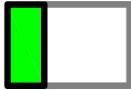
Blue

Result

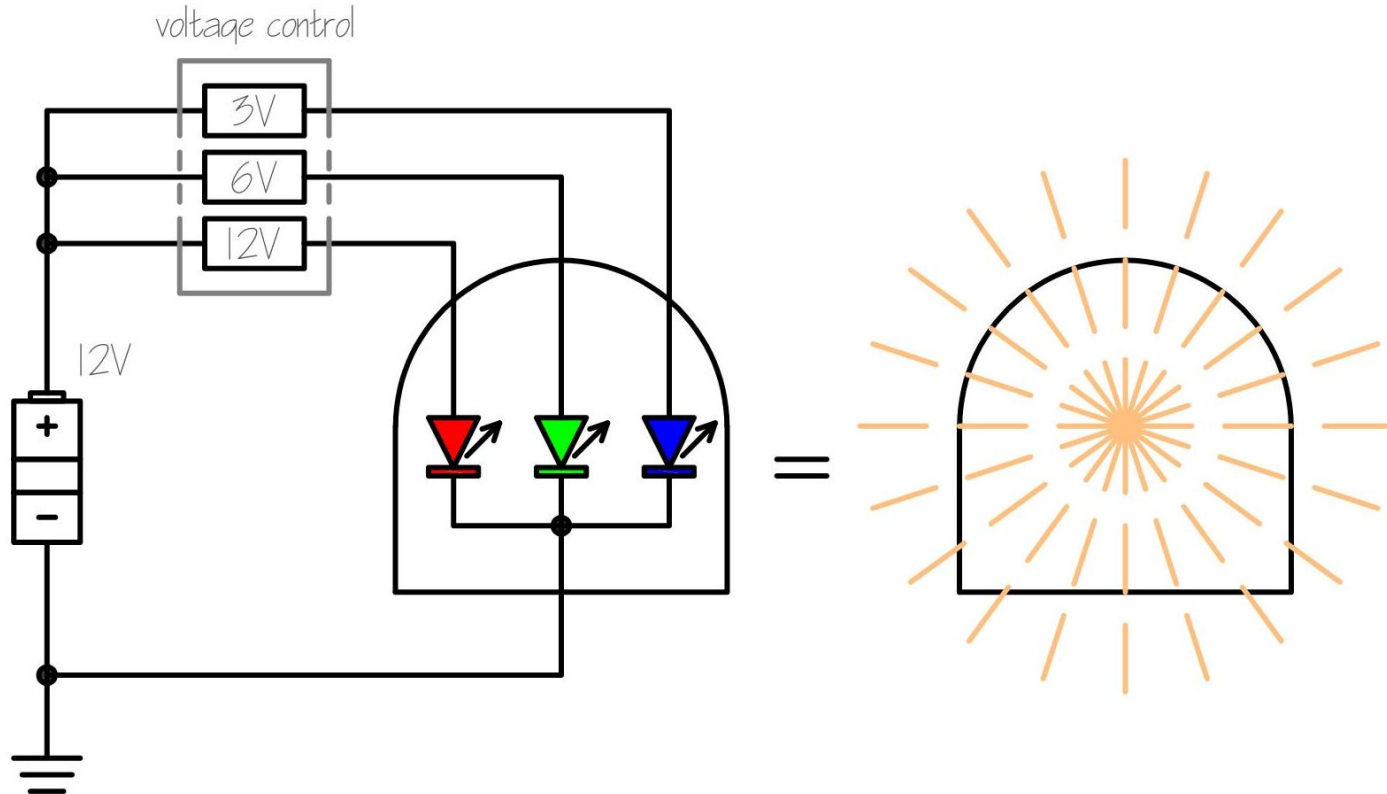


Light Primary Colors: Red, Green, Blue



Red		Green		Blue		Result
	+		+		=	
	+		+		=	
	+		+		=	
	+		+		=	

Common Anode RGB LED - Light Orange = 4:2:1 R:G:B



System Overview

Top level design

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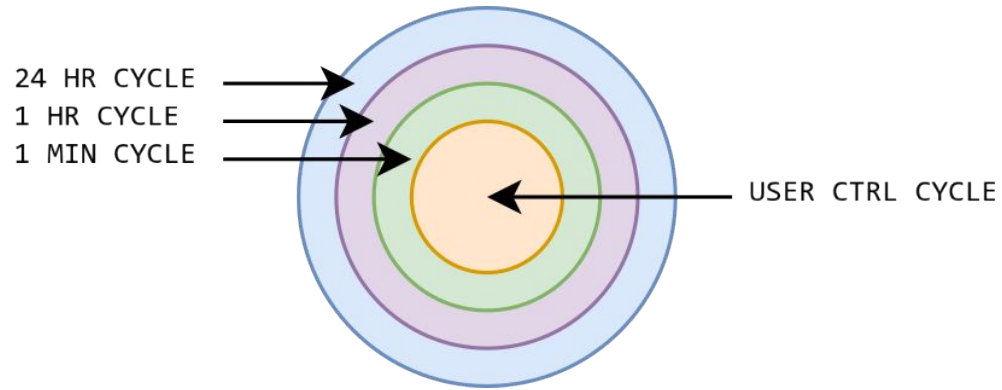
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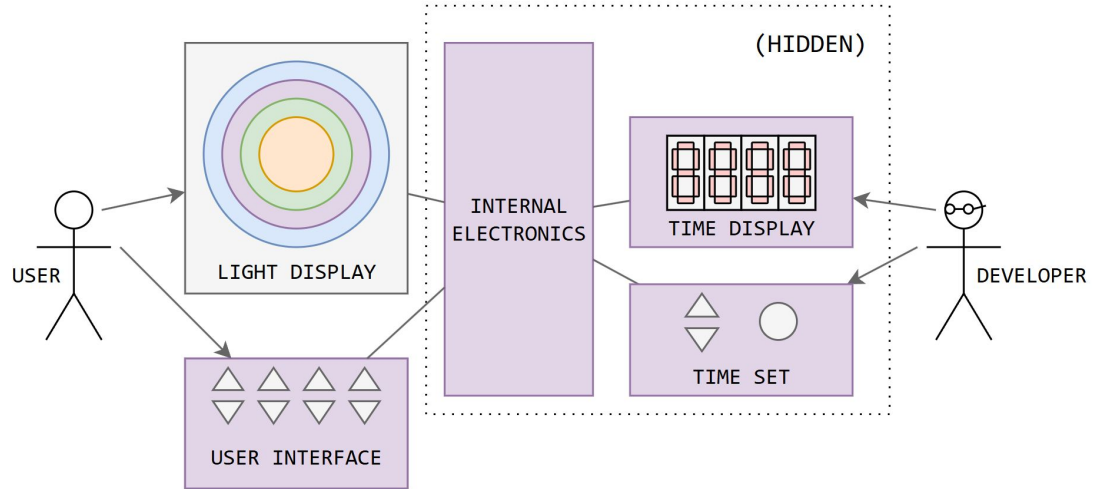
Final Thoughts

Light Display

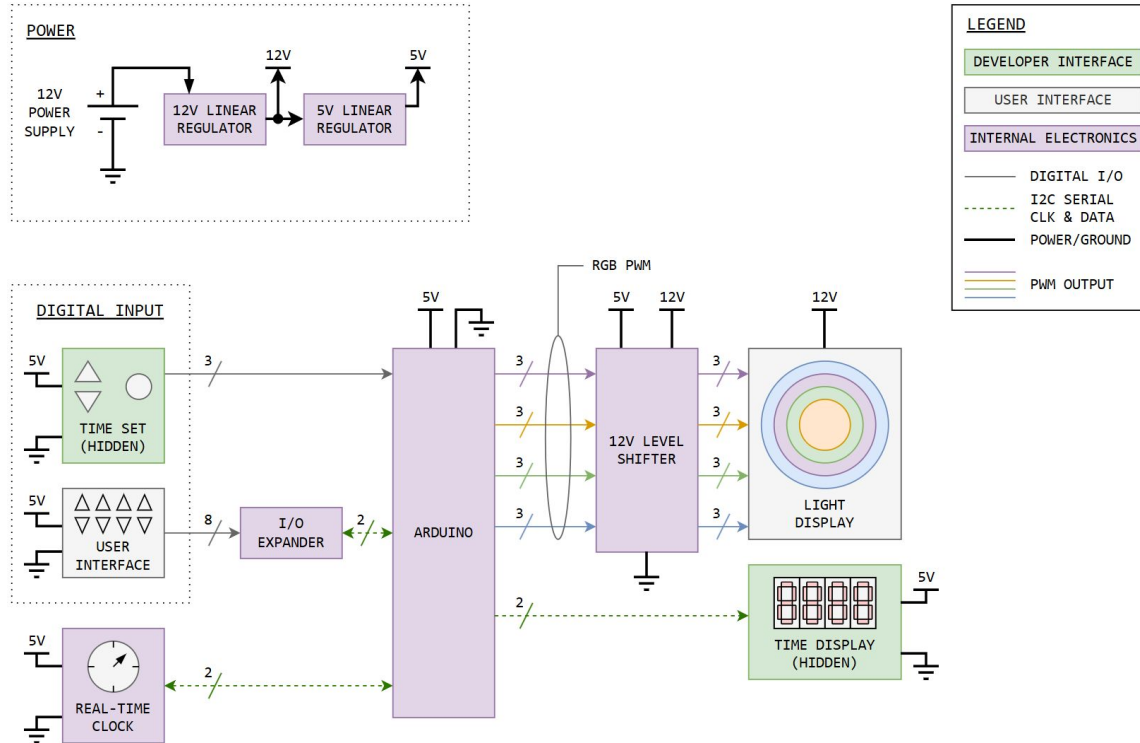


Use Case Diagram

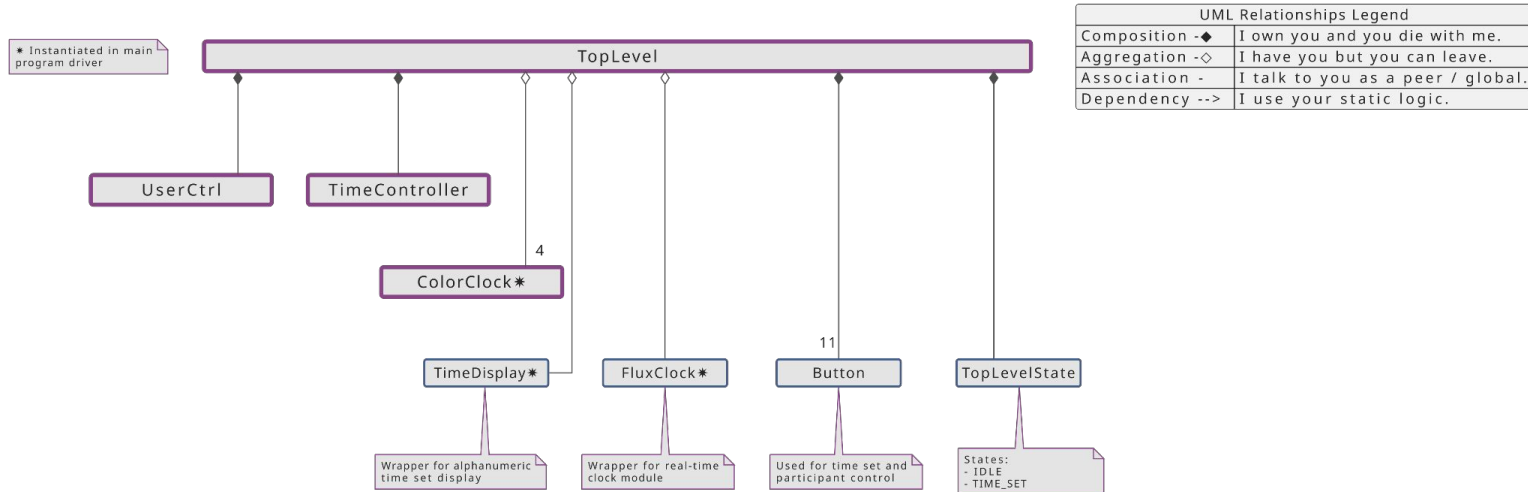
- User:
 - Views light display
 - Interacts with user control interface
- Developer:
 - Sets time
 - Views time display
- Internal electronics
 - I/O & control logic



Top Level Hardware Design

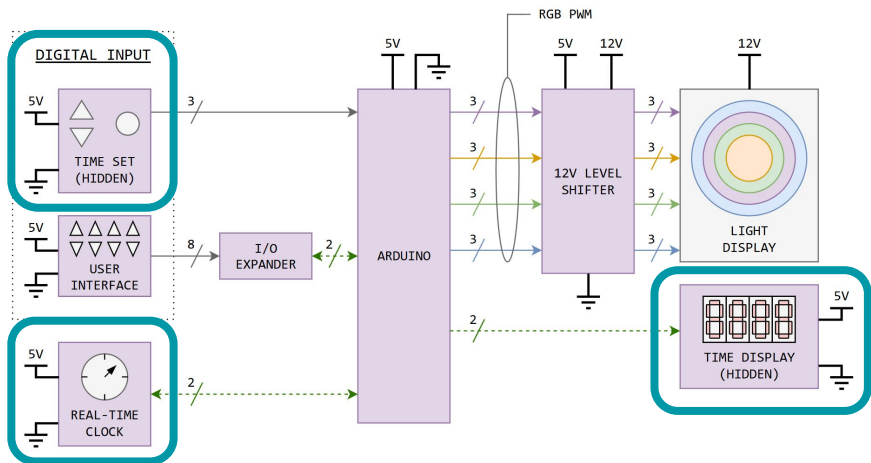


Top Level Software Architecture



Time Controller

Manual time set



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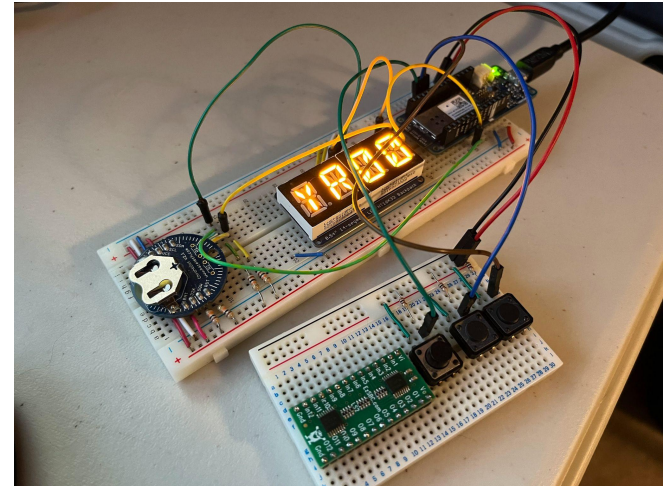
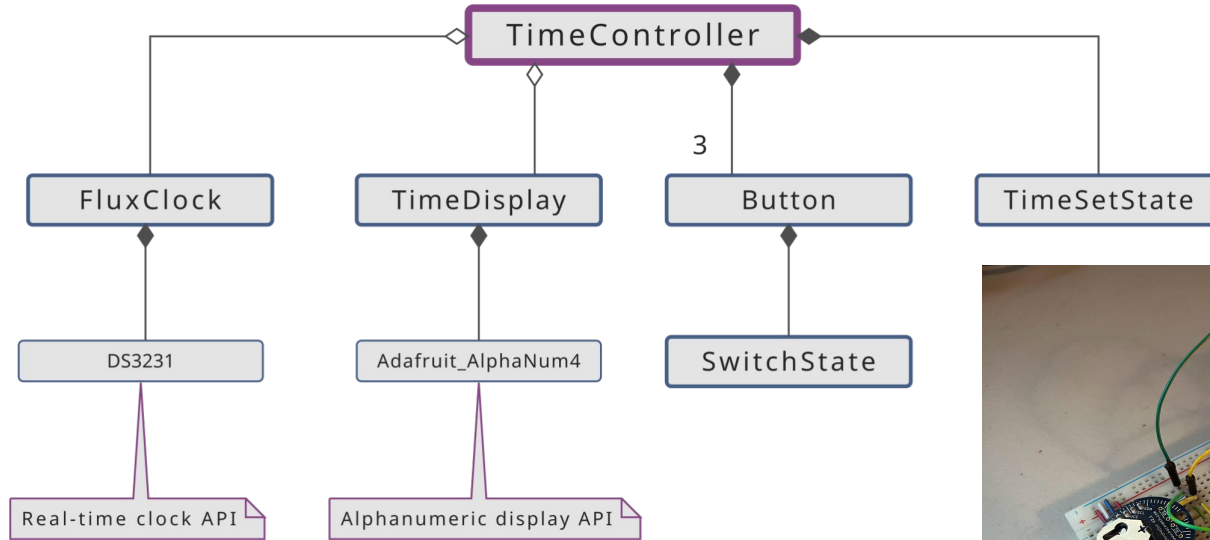
User Interface

Circuit Board Evolution

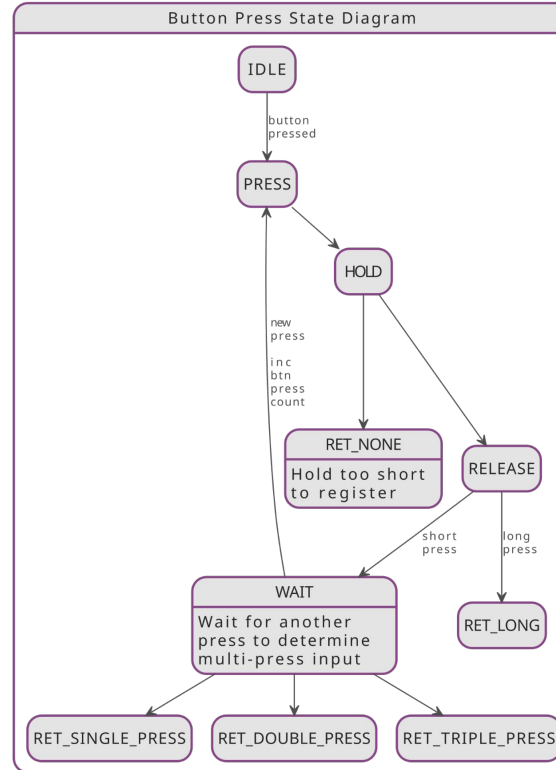
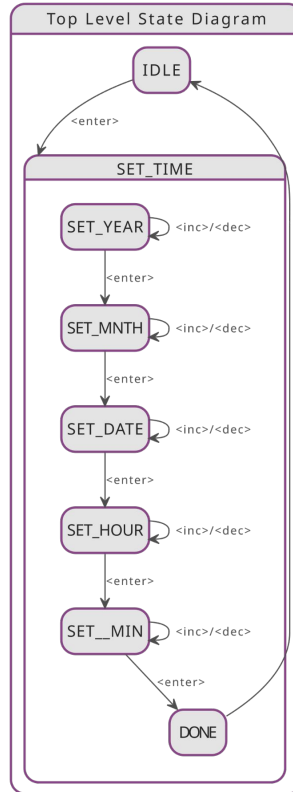
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Time Controller Architecture



State Diagrams

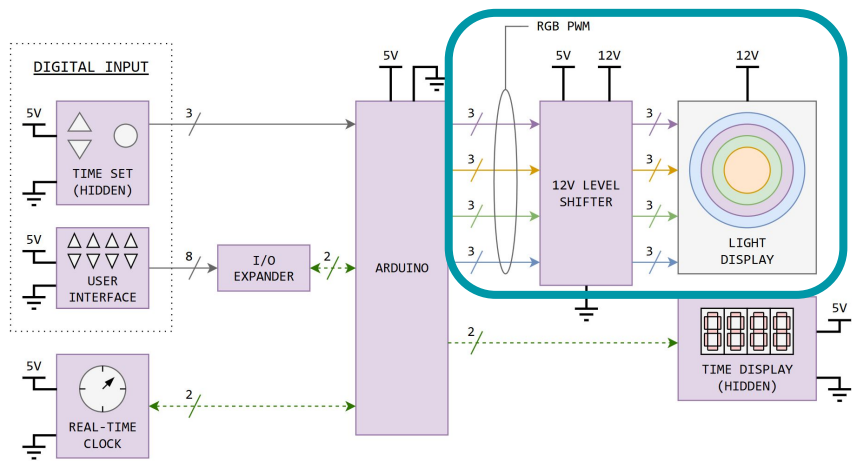


Millisecond Granularity with Clock Module + System Time

- Real-time clock module - time resolution up to **one second**
- Millisecond precision is derived by **combining**:
 - RTC time (seconds)
 - Arduino uptime (milliseconds since power-on)
- Hybrid approach enables **higher resolution** than the RTC alone
- Increased resolution -> supports smoother color transitions

Light Display

Translating time to color



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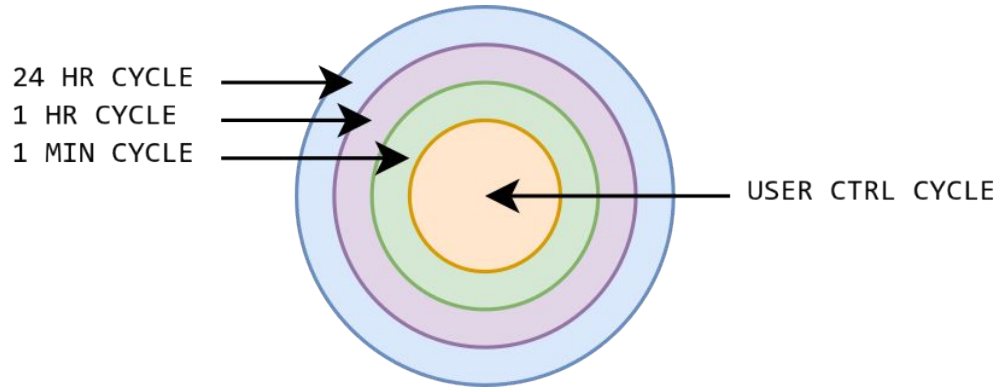
User Interface

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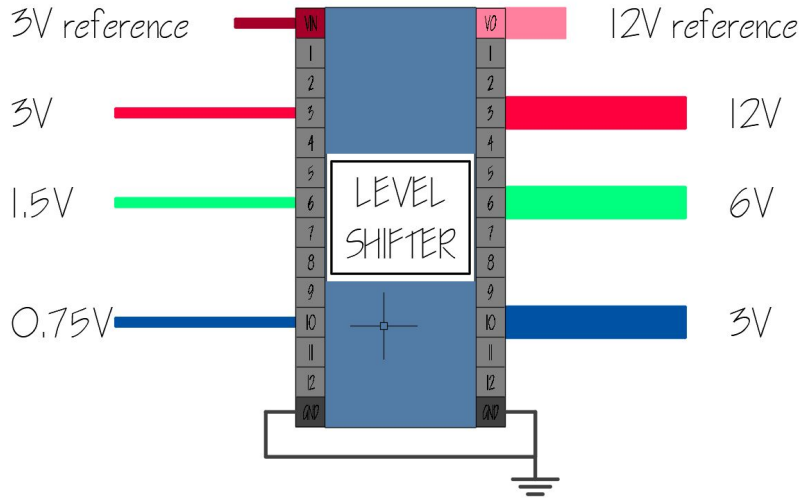
Final Thoughts

4 ColorClocks; Color Shifts at Different Frequencies



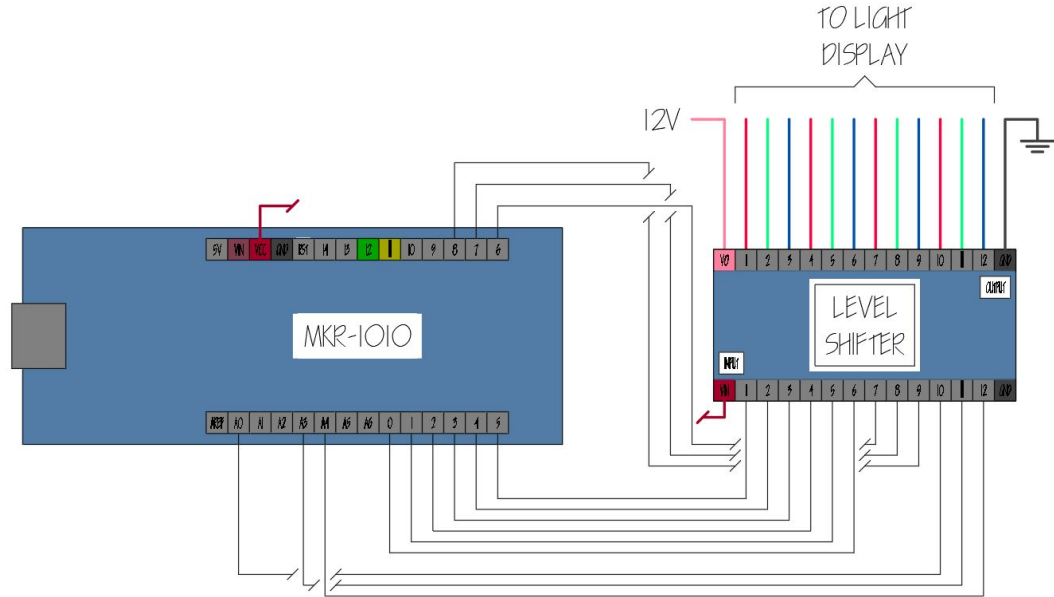
Level Shifter Scales Analog Output

- Input and output reference voltages
- Used to scale a logic level output, e.g. 3V to 12V
- Works as an amplifier in ColorClock

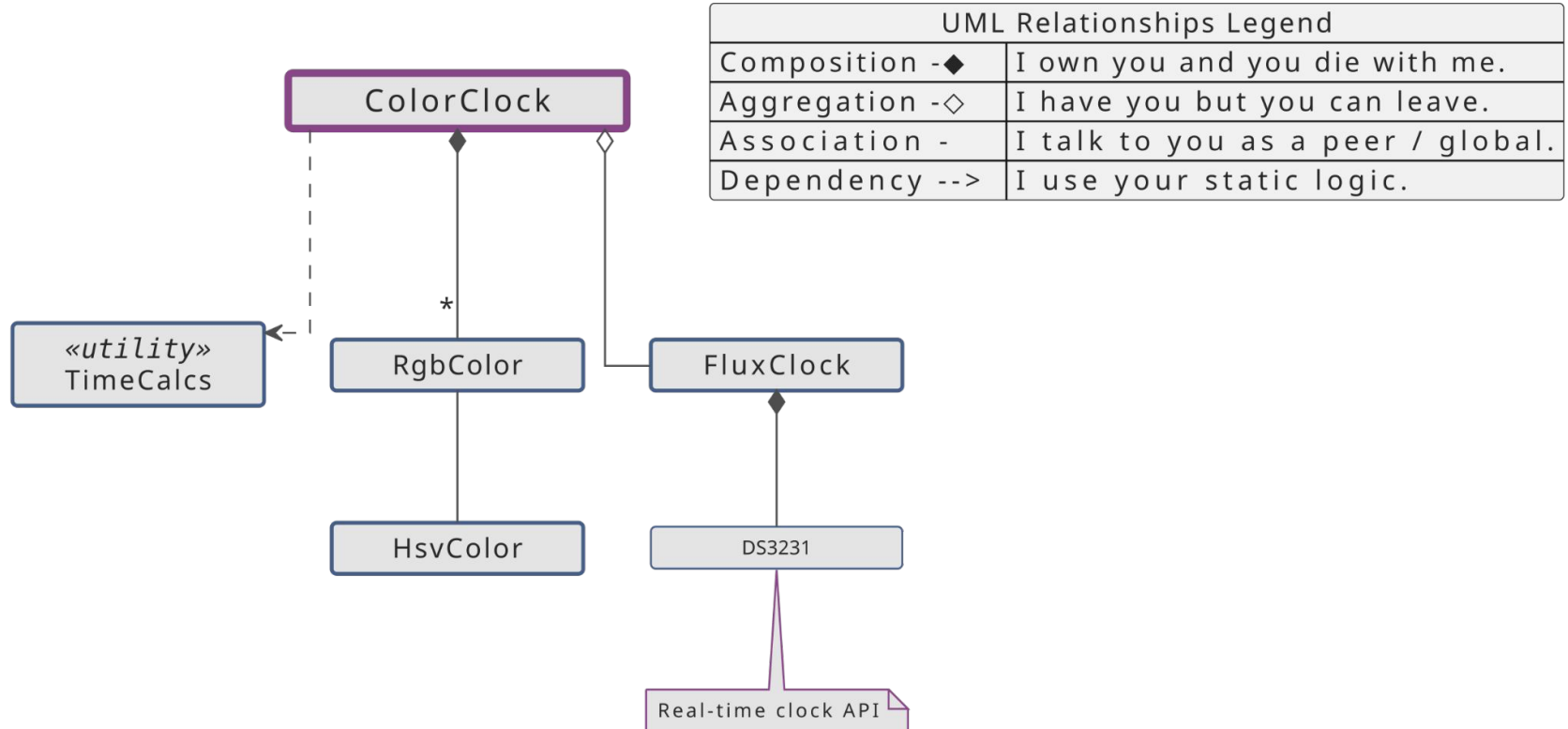


Wiring Diagram Arduino to Level Shifter

- Four ColorClock objects, each with RGB from PWM pins
- Microcontroller outputs max 5V; LEDs require 12V
- Level shifter increases output voltage from PWM output pins

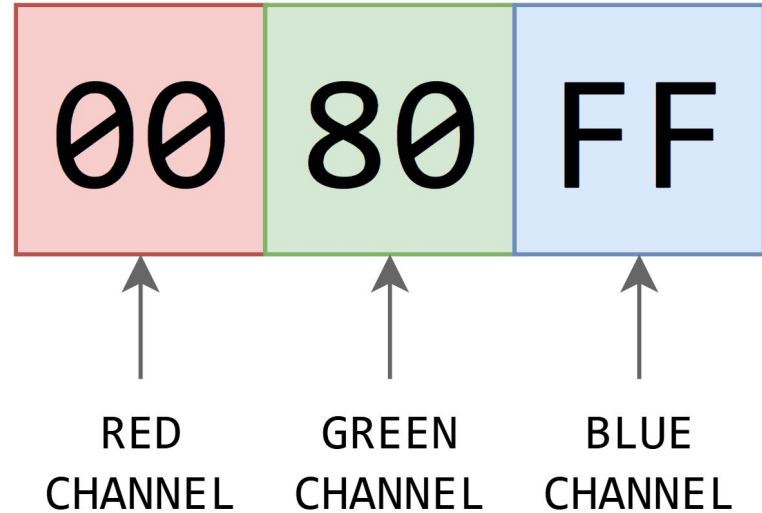


ColorClock Class Diagram



Colors Expressed as 24-Bit RGB Values

- Red, green, and blue channels use intensity values from 0–255 (0x00 → 0xFF hexadecimal)
- PWM output interprets each channel value as a **fraction** of full-scale output (value / 255)

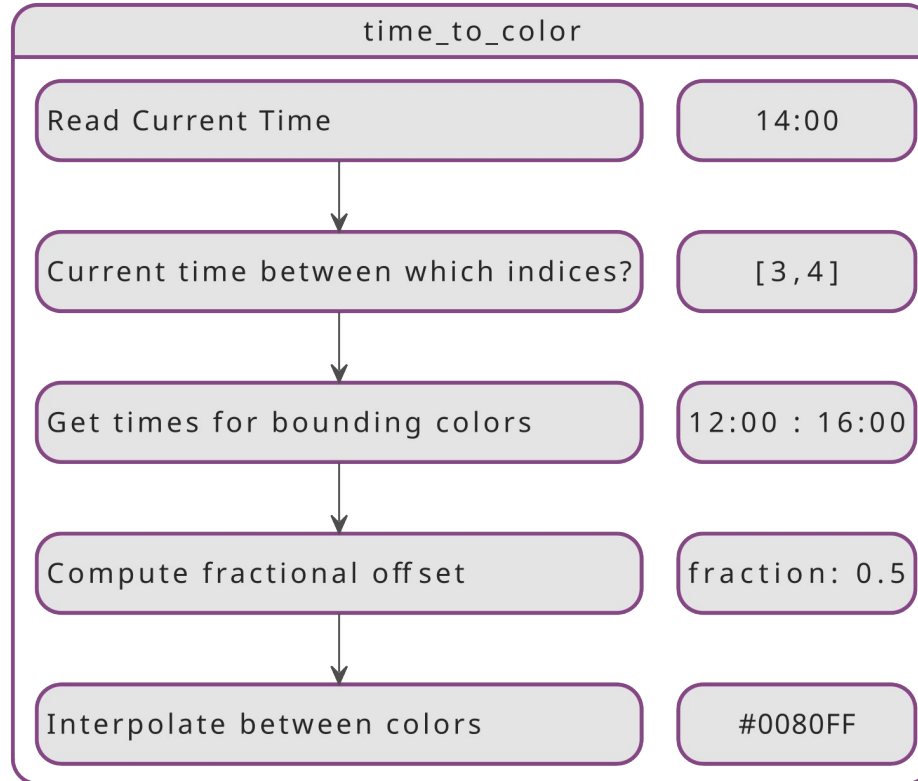


Parallel Time-Color Mapping

- Two parallel arrays, equal length
- One array stores colors
- One array stores the corresponding times for each color

color_selection	color_times
#FF0000	00:00
#FFFF00	04:00
#00FF00	08:00
#00FFFF	12:00
#0000FF	16:00
#FF00FF	20:00

Color Mapping Data Flow



Color Update Timing Testing

Color Updates Every Second

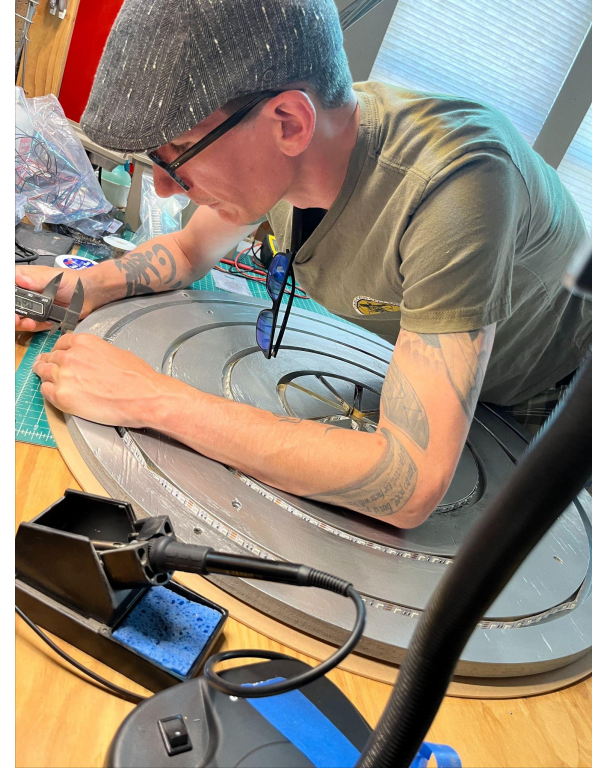
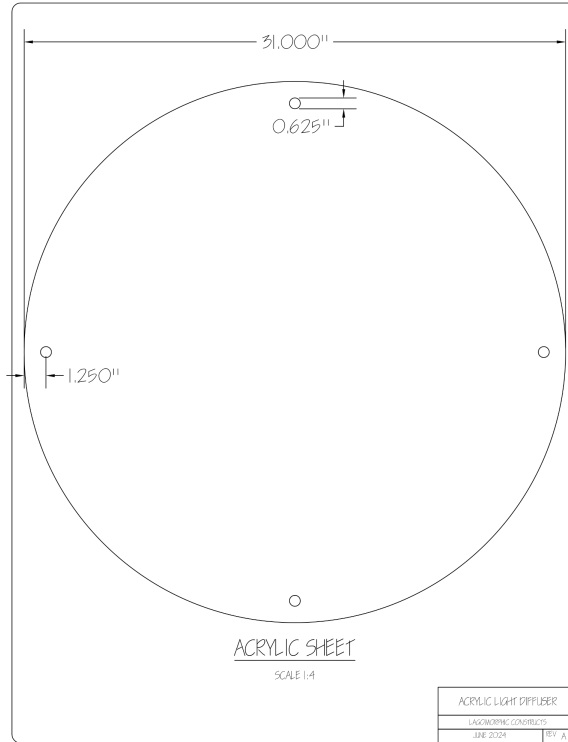


Color Updates Every Millisecond

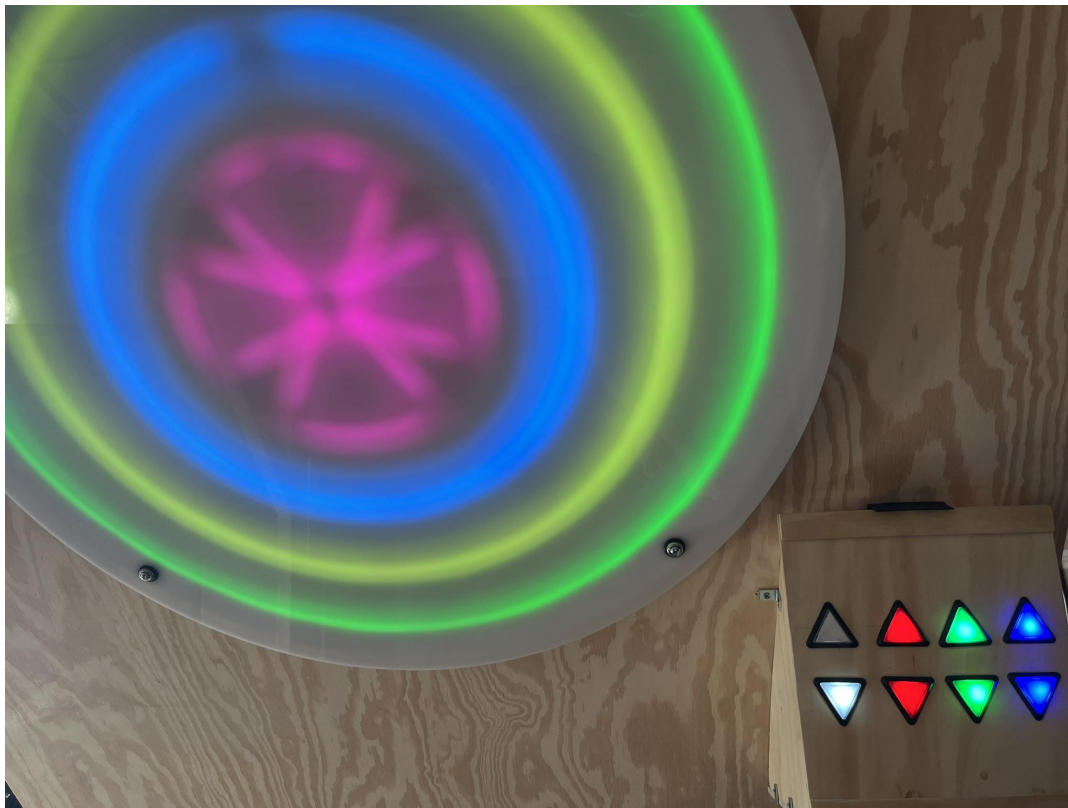


Cycle Times: Top-6s, Middle-12s, Bottom-30s

Light Display Construction

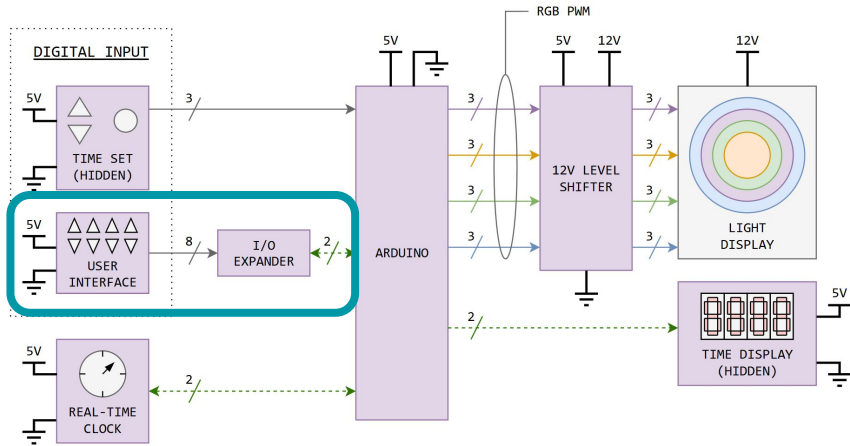


Light Display Final Implementation



User Interface

Provides interactivity



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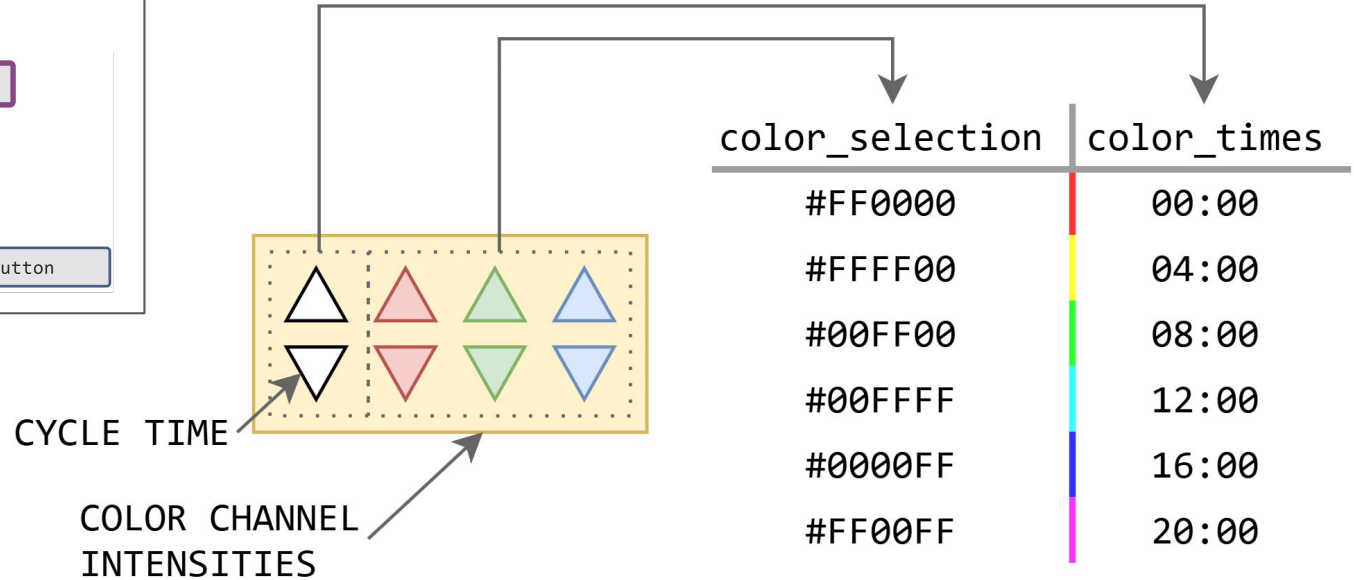
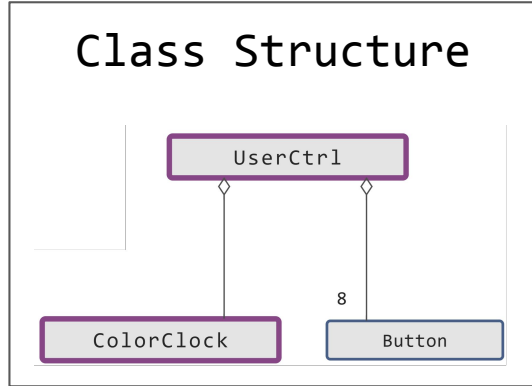
User Interface

Circuit Board Evolution

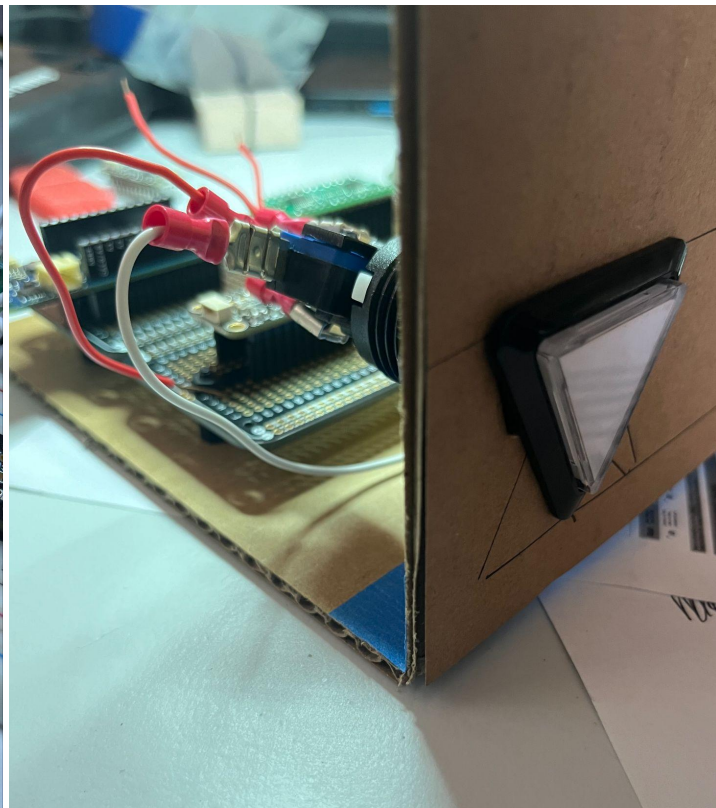
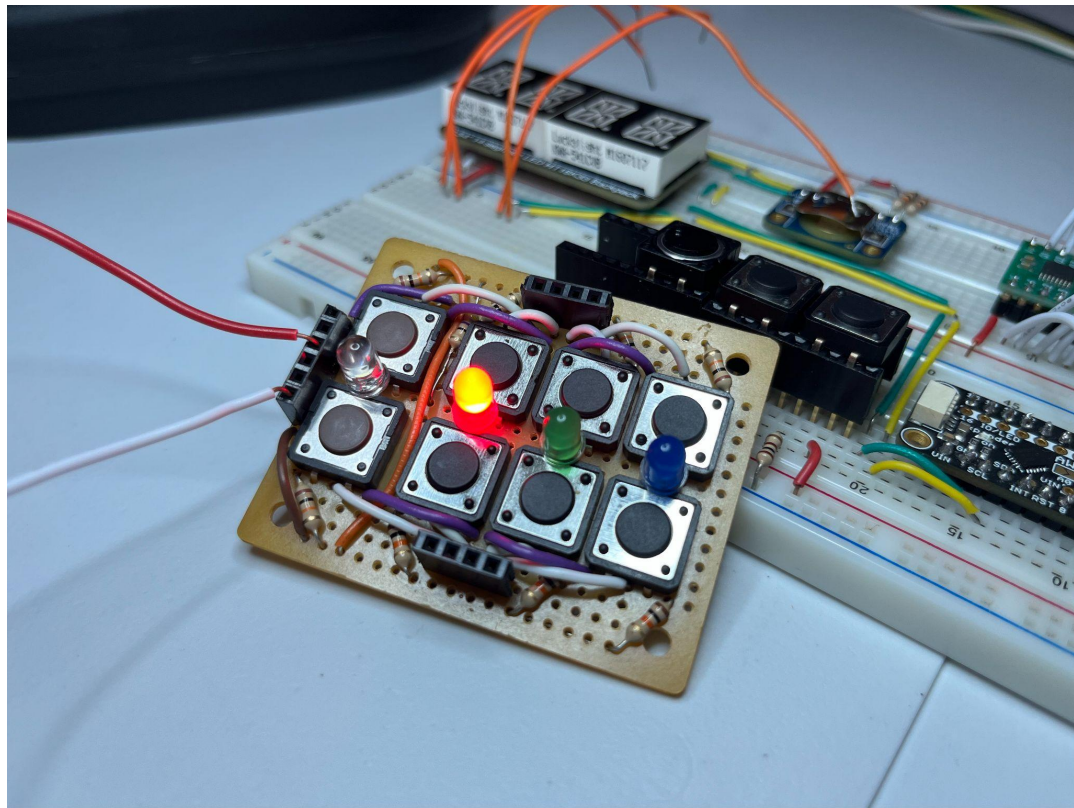
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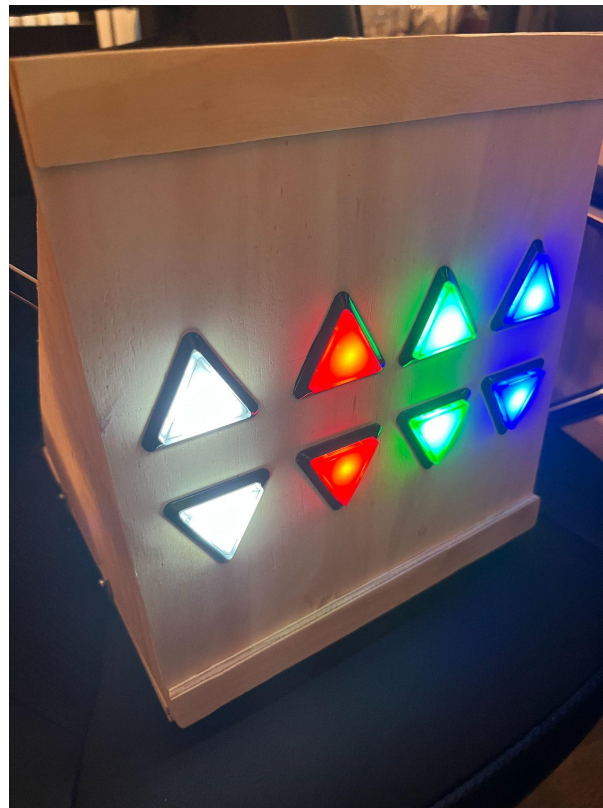
User Input Control



User Control Panel Prototyping



Final Implementation



Circuit Board Evolution

From prototype to final
implementation

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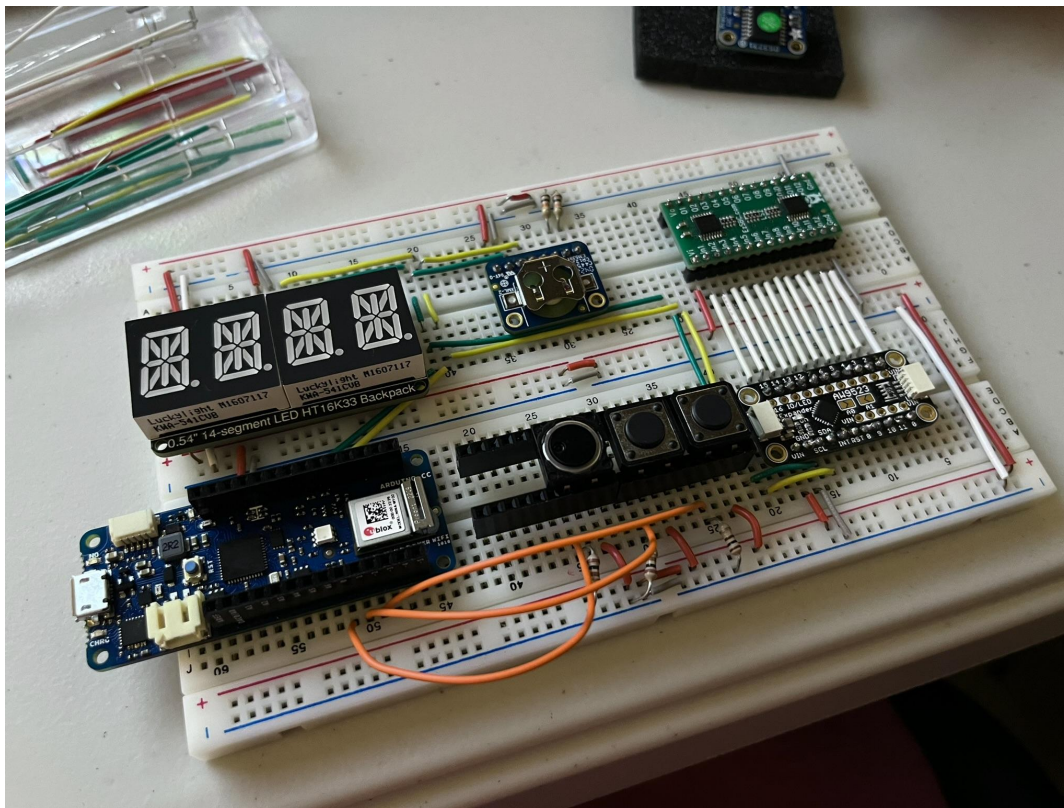
User Interface

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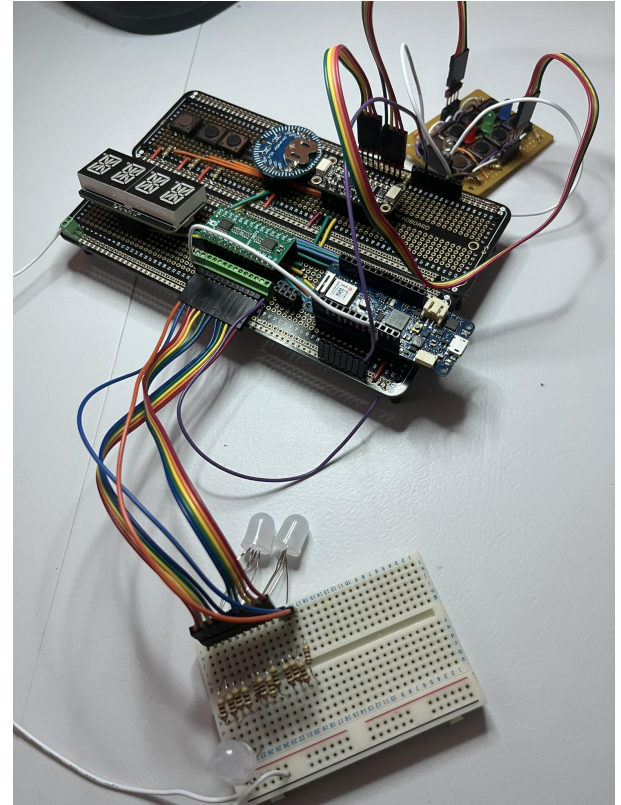
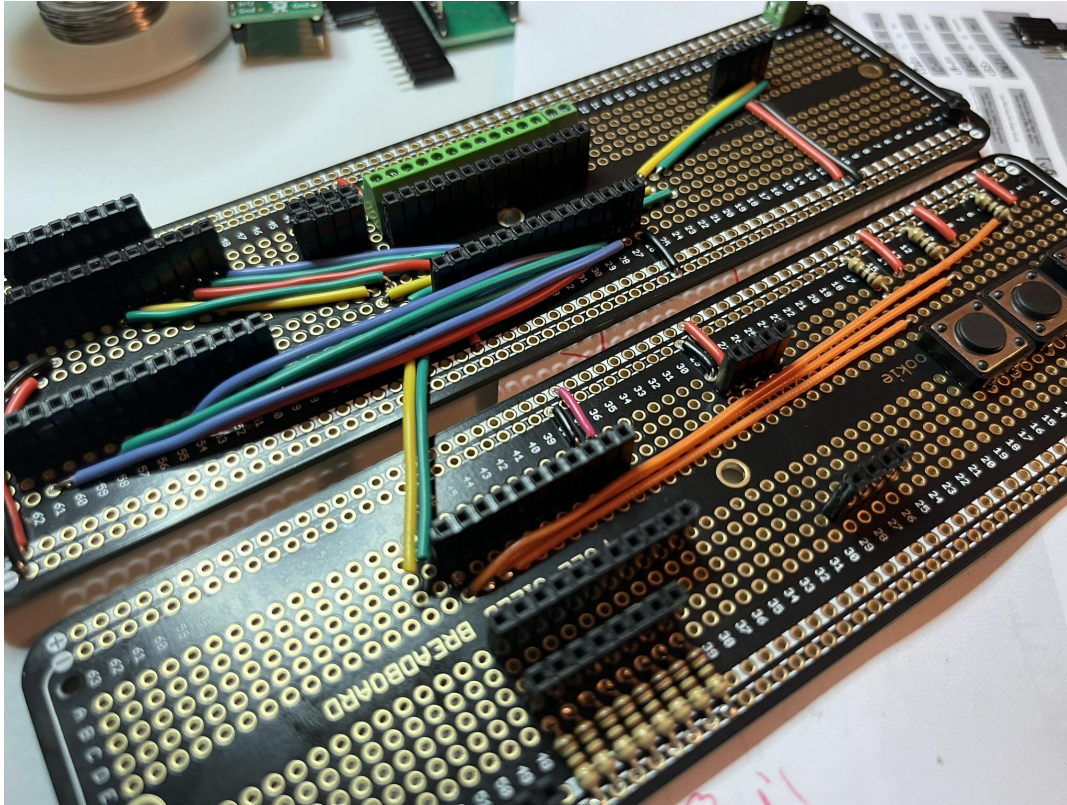
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Full Circuit Board Solderless Prototype

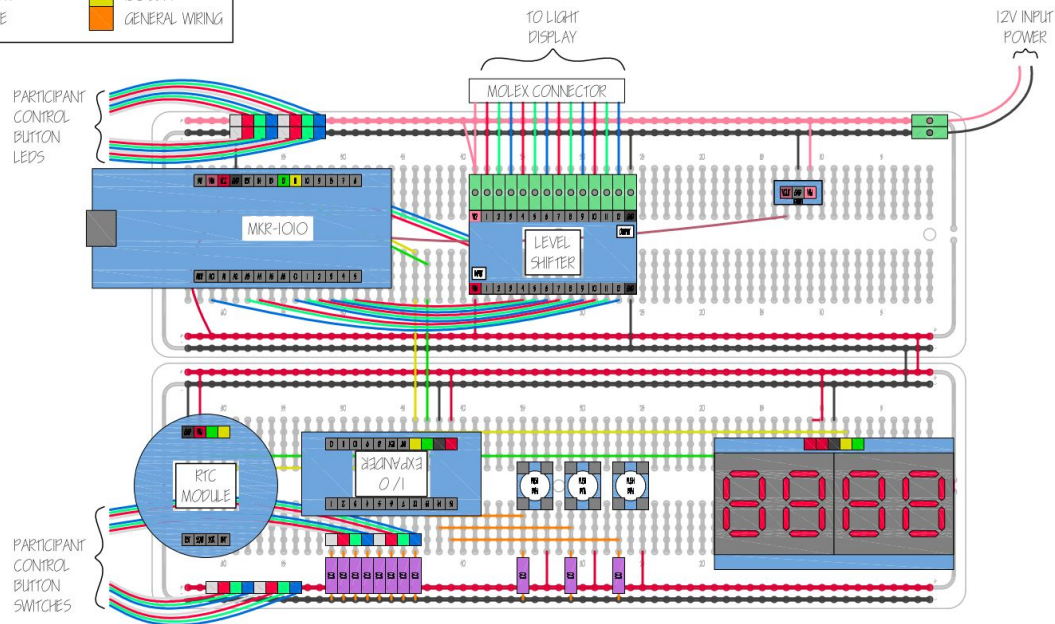


Socketed Components for Rapid Replacement



LEGEND

12V	RED LED
3.3V	GREEN LED
GND	BLUE LED
TERMINAL BLOCK	WHITE LED
GENERAL CONTACT	12C SCLK
RESISTOR	12C SDA
MODULE	GENERAL WIRING



CIRCUIT BOARD LAYOUT

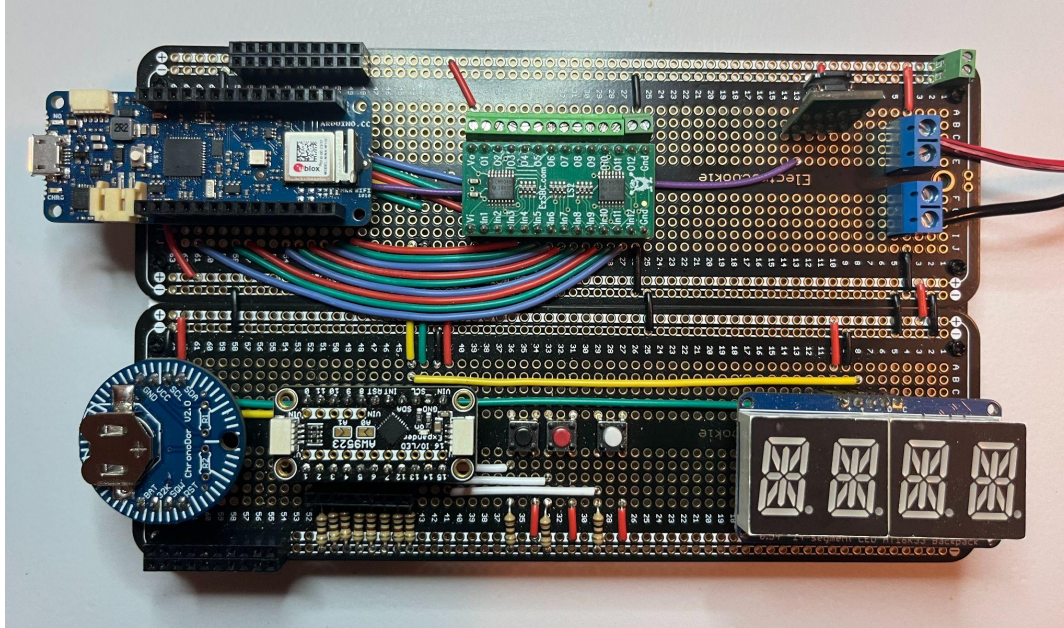
CIRCUIT BOARD LAYOUT

LACOMORPHIC CONSTRUCTS

JULY 2024

REV -

Final Implementation



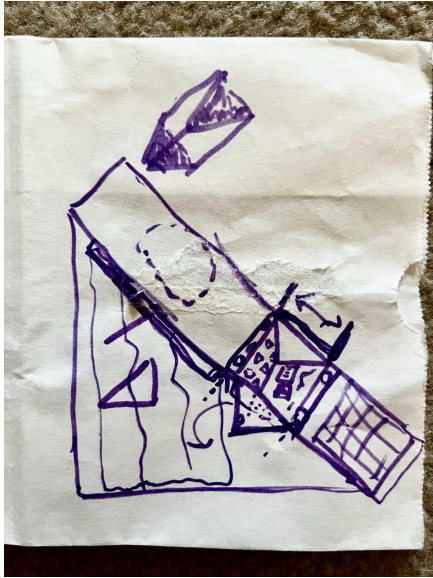
Infrastructure & Deployment

Structural implementation
and build

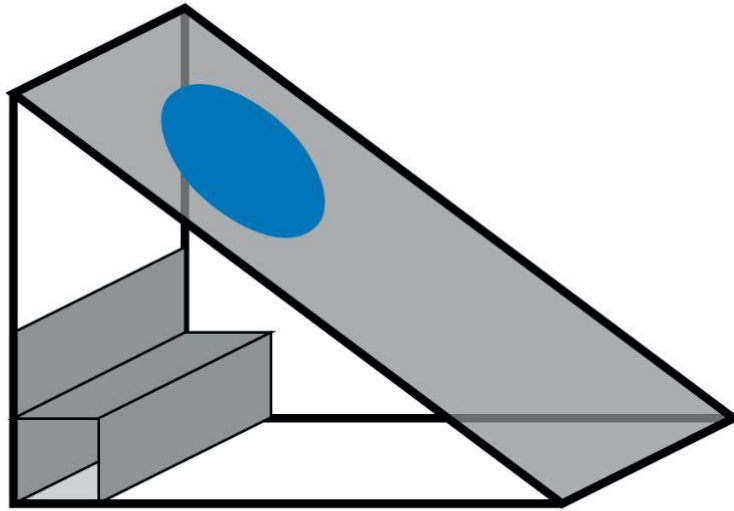
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Early Design Limitations



Napkin Sketch

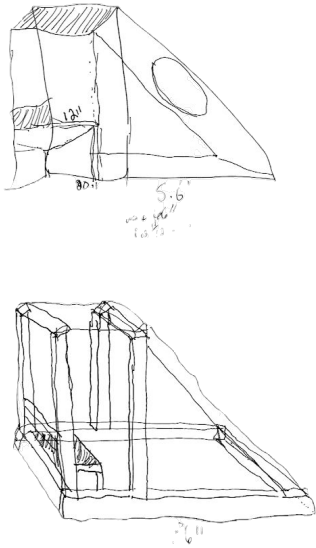


Computer Rendering

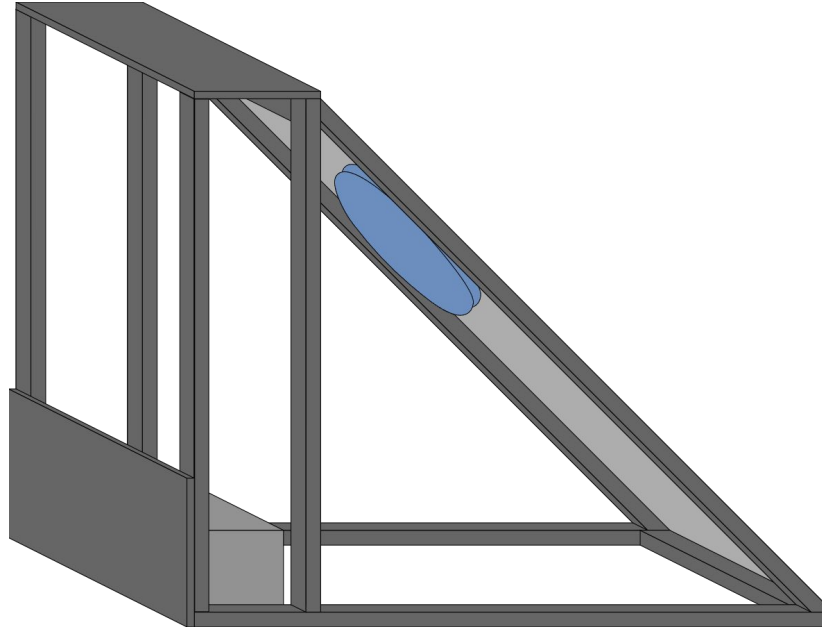


Physical
Realization

Extended Roof Design



Napkin Sketch

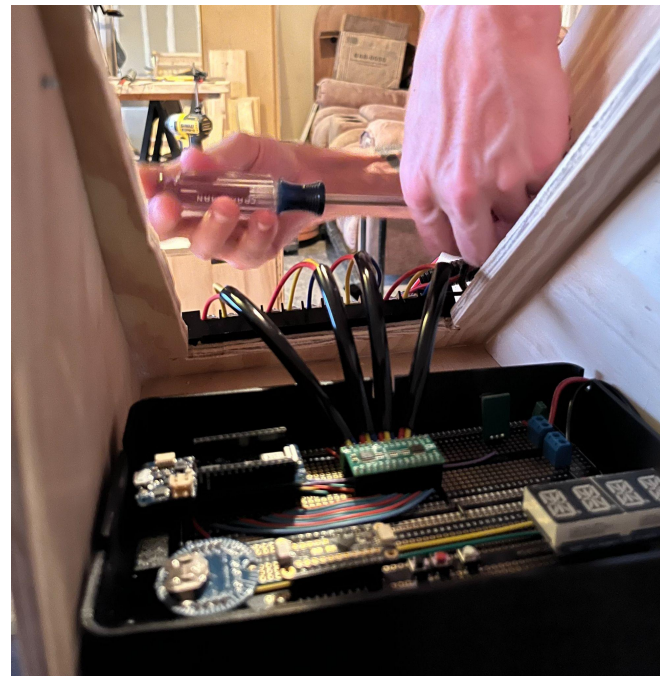


AutoCAD Rendering

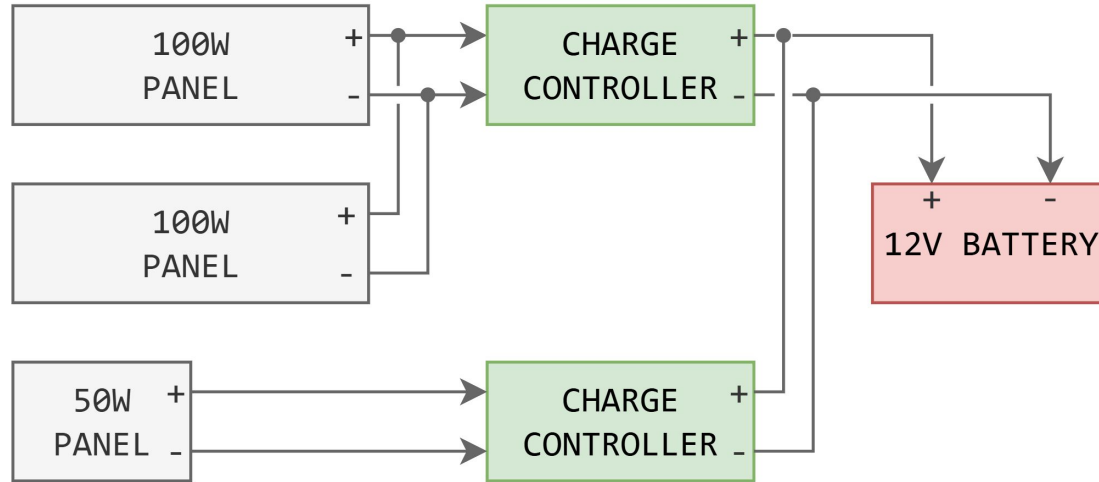


Physical
Construction

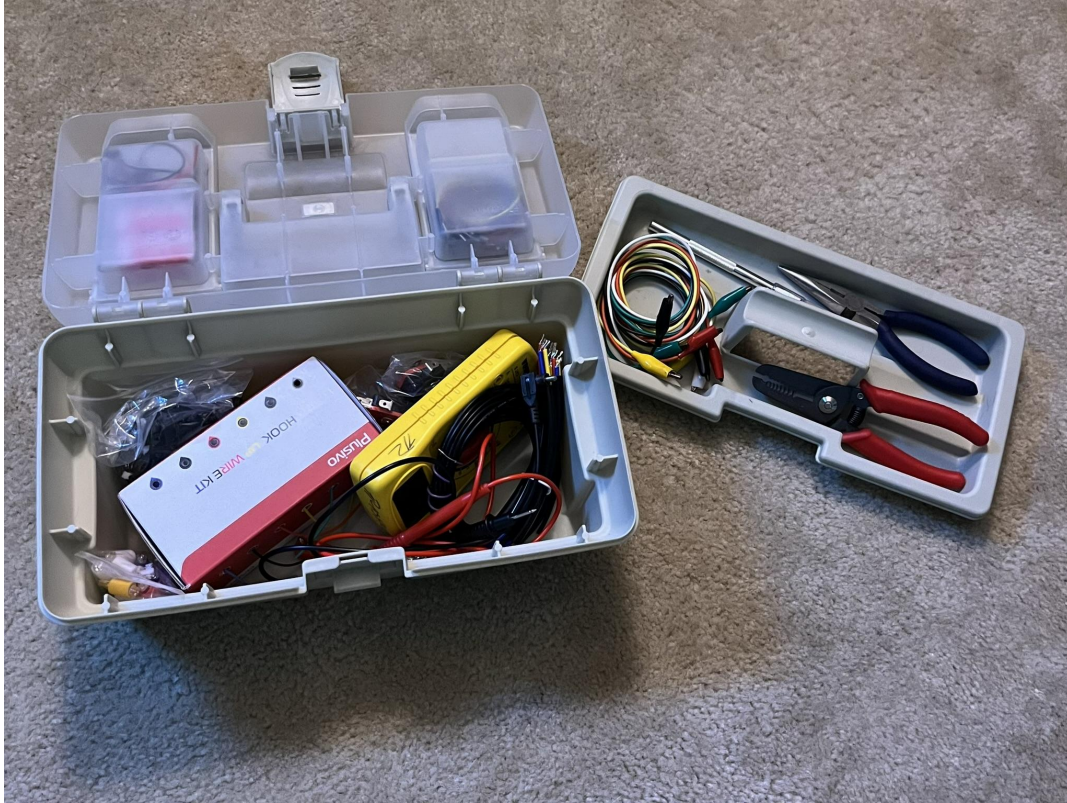
Physical Assembly



Solar Power Block Diagram



Contingency Planning



Build & On-Site Installation



Final Result



Final Thoughts

If I could do it again...

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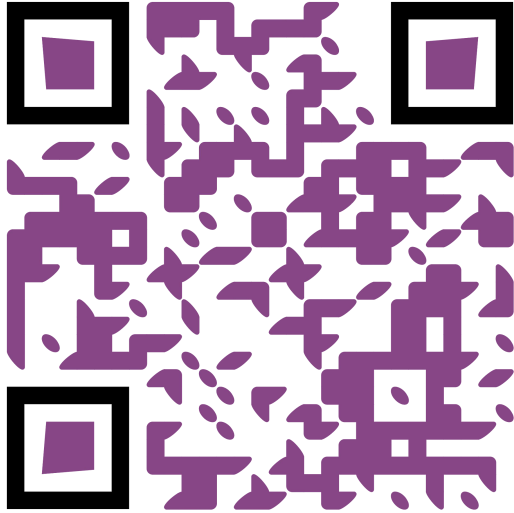
Final Thoughts

ColorClock Version 2?

Issue	Proposed Improvement
PWM made wiring challenging, not enough I/O pins	Use addressable LEDs
Hand soldering wires is tedious	Design printed circuit board
User interface confusion	More intuitive user interface - a key perhaps?
Visual presentation	Improve visual aesthetic
Limited daytime visibility	Full enclosure?

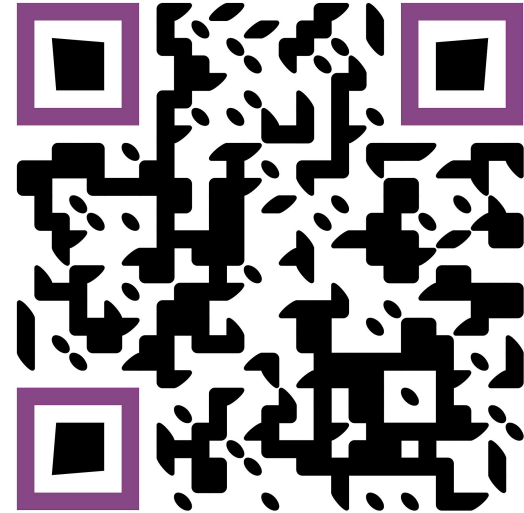


ColorClock Links



Source Code

github.com/rebeccajennifer/color-clock/tree/main/sw/arduino/color-clock-driver



Website

rebeccajennifer.com/projects/color-clock